

Unit-1

1.Bootstrap Classname

Bootstrap Class Names

♦ Definition

In Bootstrap, **class names** are predefined CSS classes provided by the framework. They are applied to HTML elements to quickly style layouts, components, and utilities without writing custom CSS.

Bootstrap works mainly through these class names.

♦ Types of Bootstrap Class Names

1) Layout Classes (Grid System)

Used to create responsive layouts.

Container Classes

```
<div class="container"></div>  
<div class="container-fluid"></div>
```

- **container** → Fixed width
- **container-fluid** → Full width

Row Class

```
<div class="row"></div>
```

Column Classes

```
<div class="col-6"></div>  
<div class="col-md-4"></div>
```

- **col-6** → 50% width
- **col-md-4** → 4 columns on medium screens

2) Typography Classes

```
<p class="text-center"></p>
<p class="text-danger"></p>
<p class="fw-bold"></p>
```

- `text-center` → Center aligned
- `text-danger` → Red color
- `fw-bold` → Bold text

3 Button Classes

```
<button class="btn btn-primary">Click</button>
<button class="btn btn-success">Save</button>
<button class="btn btn-danger">Delete</button>
```

- `btn` → Base button
- `btn-primary`, `btn-success`, etc. → Different styles

4 Background & Color Classes

```
<div class="bg-primary text-white"></div>
<div class="bg-warning"></div>
```

- `bg-primary` → Blue background
- `text-white` → White text

5 Spacing Classes

```
<div class="m-3"></div>
<div class="p-4"></div>
```

- `m-3` → Margin
- `p-4` → Padding

6 Display & Utility Classes

```
<div class="d-none"></div>
<div class="d-flex"></div>
<div class="justify-content-center"></div>
```

Used for layout control.

♦ How Bootstrap Class Names Work

- Bootstrap provides predefined CSS styles.
- When you add a class name to an HTML element, the style is automatically applied.
- No need to write custom CSS.
- Responsive classes work using built-in media queries.

Short Exam Answer

Bootstrap class names are predefined CSS classes used to style and design responsive web pages. They include layout classes like container, row, and col; component classes like btn; and utility classes for spacing, color, and alignment. Bootstrap works by applying these class names to HTML elements to achieve styling without writing custom CSS.

2.CSS- levels of stylesheets

CSS – Levels of Stylesheets

CSS can be applied to HTML documents in **three different levels**. These levels define where and how the style rules are written.

1) Inline Stylesheet

♦ Definition:

Inline CSS is written **inside the opening tag of an HTML element** using the **style** attribute.

♦ Syntax:

```
<p style="color: red; font-size: 20px;">  
Hello World  
</p>
```

♦ Features:

- Applies only to a single element.
- Has the **highest priority** among all CSS levels.
- Useful for small or quick changes.

♦ Disadvantages:

- Not reusable.
- Makes code messy.
- Difficult to maintain for large projects.

2 Internal (Document-Level) Stylesheet

♦ Definition:

Internal CSS is written inside the `<style>` tag in the `<head>` section of the HTML document.

♦ Syntax:

```
<head>
<style>
p {
  color: blue;
  font-size: 18px;
}
</style>
</head>
```

♦ Features:

- Applies to the entire HTML document.
- Good for single-page websites.
- Helps maintain uniform style in one document.

♦ Disadvantages:

- Cannot be shared across multiple pages.

3 External Stylesheet

♦ Definition:

External CSS is written in a separate `.css` file and linked to the HTML document.

♦ Syntax (HTML File):

```
<link rel="stylesheet" href="style.css">
```

♦ Syntax (CSS File – style.css):

```
p {  
  color: green;  
  font-size: 16px;  
}
```

♦ Features:

- Can be applied to multiple HTML pages.
- Provides clean separation of HTML and CSS.
- Easy to maintain and update.
- Best method for large websites.

♦ Precedence Order (Priority)

When all three are used:

1. **Inline CSS** (Highest priority)
2. **Internal CSS**
3. **External CSS** (Lowest priority)

Short Exam Answer (5 Marks)

CSS can be applied in three levels: Inline, Internal, and External stylesheets. Inline CSS is written inside an HTML tag and applies to a single element. Internal CSS is written inside the `<style>` tag in the head section and applies to the whole document. External CSS is written in a separate file and linked to multiple HTML documents. Inline styles have the highest priority.

3.Selector Forms

CSS – Selector Forms

A **selector** specifies the HTML elements to which a particular CSS style rule applies. CSS provides different selector forms to target elements precisely.

❶ Simple (Element) Selector

- ◆ **Definition:**

Selects all elements of a specific type.

- ◆ **Syntax:**

```
h1 { color: blue; }
```

✓ Applies to all `<h1>` elements.

You can also group selectors:

```
h2, h3 { color: red; }
```

2 Class Selector

- ◆ **Definition:**

Selects elements with a specific class attribute.

- ◆ **Syntax:**

```
p.normal { color: black; }  
p.warning { color: red; }
```

- ◆ **HTML:**

```
<p class="warning">Important message</p>
```

✓ Allows different styles for the same element type.

3 Generic Selector

- ◆ **Definition:**

Applies style to any element with a given class (without specifying tag name).

- ◆ **Syntax:**

```
.sale { color: green; }
```

- ◆ **HTML:**

```
<h3 class="sale">Weekend Sale</h3>  
<p class="sale">Discount available</p>
```

✓ Works on multiple types of elements.

4 ID Selector

◆ Definition:

Selects one specific element with a unique id.

◆ Syntax:

```
#section14 { color: blue; }
```

◆ HTML:

```
<h2 id="section14">Calico Cats</h2>
```

✓ ID must be unique in a document.

5 Contextual Selectors

Used when style depends on element position in document.

(a) Descendant Selector

```
ul ol { color: red; }
```

✓ Selects `<ol>` elements inside `<ul>`.

(b) Child Selector

```
ol > li { color: blue; }
```

✓ Selects `<li>` that are direct children of `<ol>`.

6 Pseudo-Class Selectors

Apply style based on element state.

Example:

```
a:link { color: blue; }  
a:visited { color: purple; }  
a:hover { color: red; }  
h2:hover { color: green; }
```

✓ Changes style during user interaction.

7 Universal Selector

◆ Definition:

Selects all elements in a document.

◆ Syntax:

```
* { margin: 0; padding: 0; }
```

✓ Applies to every element.

📌 Short Exam Answer

CSS selector forms include:

1. Simple (Element) Selector
2. Class Selector
3. Generic Selector
4. ID Selector
5. Contextual Selector (Descendant & Child)
6. Pseudo-class Selector
7. Universal Selector

Selectors allow precise control over applying styles to HTML elements.

4.progress bar, grid systems

1 Progress Bar (Bootstrap)

◆ Definition

A **progress bar** is a visual indicator used to show the progress of a task (like file upload, loading process, completion percentage).

Bootstrap provides ready-made classes to create responsive progress bars.

♦ Basic Structure

```
<div class="progress">
  <div class="progress-bar" style="width: 50%;">
    50%
  </div>
</div>
```

♦ Explanation:

- **progress** → Container of progress bar
- **progress-bar** → Actual moving bar
- **width** → Defines completion percentage

♦ Colored Progress Bars

```
<div class="progress">
  <div class="progress-bar bg-success" style="width: 75%;">
    75%
  </div>
</div>
```

Common color classes:

- **bg-success**
- **bg-danger**
- **bg-warning**
- **bg-info**

♦ Striped Progress Bar

```
<div class="progress">
  <div class="progress-bar progress-bar-striped" style="width: 60%;">
    </div>
</div>
```

♦ Animated Progress Bar

```
<div class="progress">
  <div class="progress-bar progress-bar-striped progress-bar-animated"
style="width: 80%;">
  </div>
</div>
```

Uses of Progress Bar

- File uploads
- Form completion status
- Loading indicators
- Task tracking

Grid System (Bootstrap)

♦ Definition

The **Grid System** is a layout system in Bootstrap that divides the webpage into **12 equal columns**.

It helps create responsive layouts.

♦ Basic Structure

```
<div class="container">
  <div class="row">
    <div class="col-6">Column 1</div>
    <div class="col-6">Column 2</div>
  </div>
</div>
```

♦ Explanation:

- **container** → Main wrapper
- **row** → Horizontal group
- **col-6** → 6 out of 12 columns (50% width)

♦ Responsive Breakpoints

Bootstrap uses different classes for different screen sizes:

- **col-sm-** → Small devices
- **col-md-** → Medium devices

- **col-lg** → Large devices
- **col-xl** → Extra large devices

Example:

```
<div class="col-md-4 col-sm-12">
```

- On medium screen → 4 columns
- On small screen → Full width

♦ Example of 3 Equal Columns

```
<div class="row">
  <div class="col-md-4">Column 1</div>
  <div class="col-md-4">Column 2</div>
  <div class="col-md-4">Column 3</div>
</div>
```

4 + 4 + 4 = 12 columns

Short Exam Answer

- A **Progress Bar** in Bootstrap is used to visually represent task completion using **progress** and **progress-bar** classes.
- The **Grid System** divides the screen into 12 columns using container, row, and col classes to create responsive layouts.

Unit 2

1. Pattern Matching

Pattern Matching

♦ Definition

Pattern matching is the process of searching a text to find sequences of characters that match a specified pattern.

It is commonly used for:

- Validating input (email, phone number, password)

- Searching and replacing text
- Extracting specific data from strings

Pattern matching is usually implemented using **Regular Expressions (Regex)**.

♦ Regular Expressions (Regex)

A **regular expression** is a sequence of special characters that defines a search pattern.

Basic Symbols Used in Pattern Matching

Symbol	Meaning
.	Matches any single character
*	Matches 0 or more occurrences
+	Matches 1 or more occurrences
?	Matches 0 or 1 occurrence
^	Beginning of string
\$	End of string
[abc]	Matches a, b, or c
[^abc]	Matches any character except a, b, or c
[0-9]	Matches digits
[a-z]	Matches lowercase letters

♦ Example of Pattern Matching

1 Matching an Email Pattern

```
^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$
```

✓ Ensures valid email format.

♦ Pattern Matching in PHP

PHP provides built-in functions for pattern matching using regular expressions.

Common Functions:

1. `preg_match()` → Checks if pattern exists
2. `preg_replace()` → Replaces matched pattern
3. `preg_split()` → Splits string using pattern

Example in PHP

```
<?php
$email = "test@gmail.com";

if (preg_match("/^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/", $email)) {
    echo "Valid Email";
} else {
    echo "Invalid Email";
}
?>
```

✓ Checks whether the email matches the pattern.

♦ Applications of Pattern Matching

1. Form validation
2. Password strength checking
3. Data extraction
4. Log file analysis
5. Search and replace operations

📌 Short Exam Answer (4–5 Marks)

Pattern matching is the technique of searching text to find sequences that match a specific pattern. It is usually implemented using regular expressions. In PHP, pattern matching is done using functions like `preg_match()` and `preg_replace()`. It is mainly used for input validation and text processing.

2. origin of php

Origin of PHP

♦ Introduction

PHP is a **server-side scripting language** mainly used for developing dynamic web applications.

♦ Who Developed PHP?

PHP was developed by Rasmus Lerdorf in 1994.

♦ Why Was PHP Created?

- Initially created to track visitors to Lerdorf's personal website.
- It started as a set of Common Gateway Interface (CGI) programs written in C.
- The original purpose was simple web page tracking and form processing.

♦ Meaning of PHP

- Originally, PHP stood for **Personal Home Page**.
- Later, it was renamed to the recursive acronym:
PHP: Hypertext Preprocessor

♦ Growth and Evolution

- As more developers started using it, PHP evolved into a full-fledged scripting language.
- It became widely used for:
 - Form handling
 - Database access
 - Server-side web development
- It supports many databases and protocols such as POP3 and IMAP.
- It is commonly used with Apache Web Server.

♦ Importance of PHP

- Open-source and free.
- Easy to embed within HTML.
- Powerful for dynamic web applications.
- Large community support.

Short Exam Answer (5 Marks)

PHP was developed by Rasmus Lerdorf in 1994. Initially, it stood for Personal Home Page and was created to track visitors to his website. Later, it was renamed PHP: Hypertext Preprocessor and evolved into a powerful server-side scripting language used for web development, database access, and form handling.

3.form handling with example

Form Handling in PHP

◆ Definition

Form handling in PHP refers to the process of collecting user input from an HTML form and processing it on the server.

When the user clicks the **Submit** button:

1. The form data is sent to the server.
2. PHP script receives the data.
3. The script processes and returns a response.

◆ Steps in Form Handling

- 1>Create an HTML form
- 2)Use **method** (GET or POST)
- 3)Specify **action** (PHP file)
- 4)Access form data using **\$_GET** or **\$_POST**

◆ Example of Form Handling (Using POST)

Step 1: HTML Form

```
<!DOCTYPE html>
<html>
<body>

<form method="post" action="process.php">
  Name: <input type="text" name="username"><br><br>
  Age: <input type="text" name="age"><br><br>
  <input type="submit" value="Submit">
</form>

</body>
</html>
```

Step 2: PHP File (process.php)

```
<?php
$name = $_POST["username"];
```

```
$age = $_POST["age"];
```

```
echo "Hello " . $name . "<br>";  
echo "Your age is " . $age;  
?>
```

♦ Explanation

- `method="post"` → Sends data securely (not visible in URL).
- `action="process.php"` → Specifies PHP file to handle form.
- `$_POST[]` → Superglobal array used to access POST data.

♦ GET Method Example

```
<form method="get" action="process.php">
```

Then in PHP:

```
$name = $_GET["username"];
```

♦ Difference Between GET and POST

GET	POST
Data visible in URL	Data hidden
Limited size	Large data allowed
Less secure	More secure

📌 Short Exam Answer (5 Marks)

Form handling in PHP is the process of collecting and processing user input from an HTML form. The form uses GET or POST methods to send data to a PHP script specified in the action attribute. The PHP script retrieves the data using `$_GET` or `$_POST` arrays and processes it accordingly.

4.array

Array in PHP

♦ Definition

An **array** in PHP is a data structure used to store multiple values in a single variable.

Unlike many programming languages, PHP arrays are flexible and store data in the form of **key–value pairs**.

♦ **Types of Arrays in PHP**

PHP supports three main types of arrays:

1 Indexed (Numeric) Array

- Uses numeric indexes (0, 1, 2, ...).
- Index starts from 0 by default.

♦ **Example:**

```
<?php
$numbers = array(10, 20, 30, 40);
echo $numbers[2]; // Outputs 30
?>
```

Or:

```
$numbers[0] = 10;
$numbers[1] = 20;
```

2 Associative Array

- Uses **string keys** instead of numbers.
- Useful for storing labeled data.

♦ **Example:**

```
<?php
$ages = array(
    "Ravi" => 21,
    "Meena" => 22,
    "Arjun" => 23
);

echo $ages["Meena"]; // Outputs 22
?>
```

3 Multidimensional Array

- An array containing one or more arrays inside it.

♦ Example:

```
<?php
$students = array(
    array("Ravi", 21),
    array("Meena", 22),
    array("Arjun", 23)
);

echo $students[1][0]; // Outputs Meena
?>
```

♦ Creating Arrays

Method 1: Using **array()** function

```
$list = array(1, 2, 3);
```

Method 2: Using square brackets (Modern PHP)

```
$list = [1, 2, 3];
```

♦ Important Array Functions

- **count()** → Counts elements
- **sort()** → Sorts array
- **array_push()** → Adds element
- **array_pop()** → Removes last element
- **foreach** → Iterates array

Example:

```
<?php
$colors = ["Red", "Green", "Blue"];

foreach ($colors as $color) {
    echo $color . "<br>";
}
```

?>

Short Exam Answer (5 Marks)

An array in PHP is a data structure that stores multiple values in a single variable. PHP arrays are key–value based and can be indexed, associative, or multidimensional. Arrays can be created using the `array()` function or square brackets and accessed using indexes or keys.